

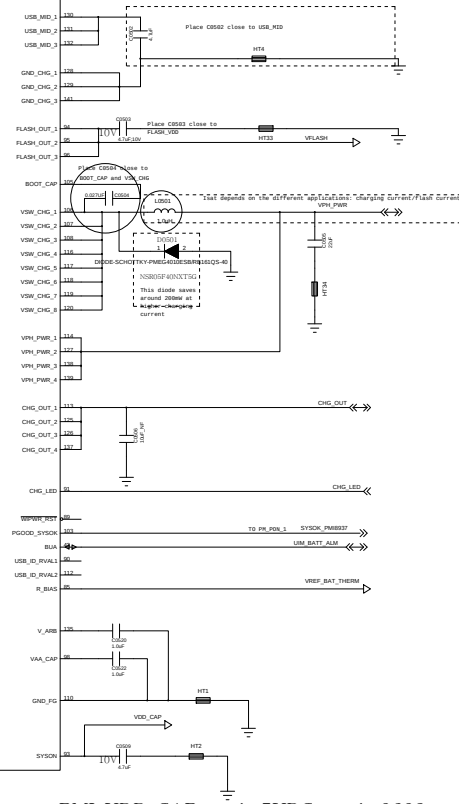
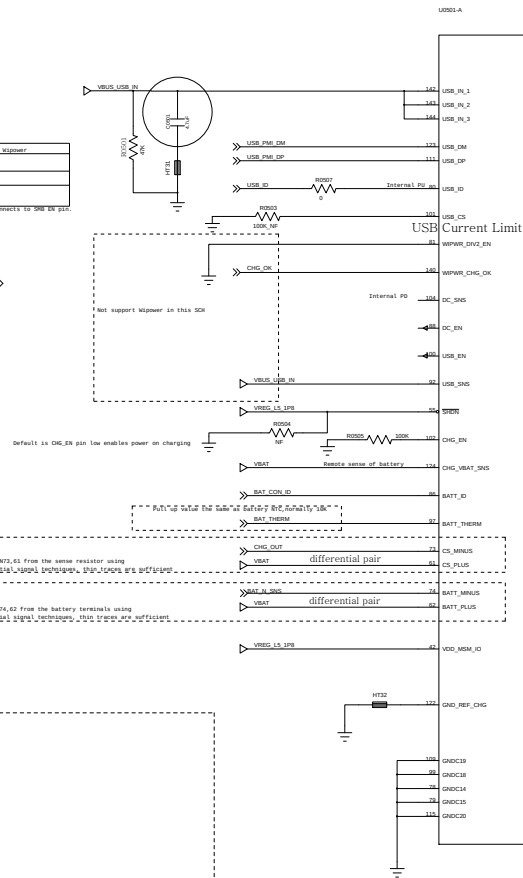
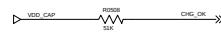
MSM8937 GPIO Configuration					
GPIO_0	NC	GPIO_41	SCAM_ID	GPIO_82	FM_DATA
GPIO_1	NC	GPIO_42	ACCL_INT1_N	GPIO_83	BT_CTL
GPIO_2	NC	GPIO_43	ALSP_INT_N	GPIO_84	BT_DATA
GPIO_3	WDOG_DISABLE	GPIO_44	MAG_RST_N	GPIO_85	NC
GPIO_4	UART_MSM_TX	GPIO_45	NC	GPIO_86	NC
GPIO_5	UART_MSM_RX	GPIO_46	NC	GPIO_87	NC
GPIO_6	NC	GPIO_47	NC	GPIO_88	NC
GPIO_7	NC	GPIO_48	NC	GPIO_89	NC
GPIO_8	NC	GPIO_49	UIM_BATT_ALM	GPIO_90	NC
GPIO_9	NC	GPIO_50	NC	GPIO_91	KEY_VOL_UP_N
GPIO_10	TP_I2C_SDA	GPIO_51	UIM1_DATA	GPIO_92	NC
GPIO_11	TP_I2C_SCL	GPIO_52	UIM1_CLK	GPIO_93	NC
GPIO_12	NC	GPIO_53	UIM1_RESET	GPIO_94	NC
GPIO_13	NC	GPIO_54	NC	GPIO_95	NC
GPIO_14	SENSOR_I2C_SDA	GPIO_55	UIM2_DATA	GPIO_96	NC
GPIO_15	SENSOR_I2C_SCL	GPIO_56	UIM2_CLK	GPIO_97	NC
GPIO_16	NC	GPIO_57	UIM2_RESET	GPIO_98	NC
GPIO_17	NC	GPIO_58	NC	GPIO_99	NC
GPIO_18	NC	GPIO_59	NC	GPIO_100	RFFE1_CLK
GPIO_19	NC	GPIO_60	LCD_RST_N	GPIO_101	RFFE1_DATA
GPIO_20	ACC_SPI_MOSI	GPIO_61	LCD_ID	GPIO_102	NC
GPIO_21	ACC_SPI_MISO	GPIO_62	NC	GPIO_103	NC
GPIO_22	ACC_SPI_CS	GPIO_63	APP_BOOT_FROM_ROM	GPIO_104	NC
GPIO_23	ACC_SPI_CLK	GPIO_64	TP_RST_N	GPIO_105	GRFC1_SEL
GPIO_24	LCD_TE0	GPIO_65	TP_INT_N	GPIO_106	NC
GPIO_25	NC	GPIO_66	GP_PDM_A0	GPIO_107	GRFC3_SEL
GPIO_26	CAM_MCLK0	GPIO_67	SDCARD_DET_N	GPIO_108	NC
GPIO_27	NC	GPIO_68	NC	GPIO_109	GRFC5_SEL
GPIO_28	CAM_MCLK2	GPIO_69	CDC_PDM_CLK	GPIO_110	GRFC6_SEL
GPIO_29	CAM_I2C_SDA0	GPIO_70	CDC_PDM_SYNC	GPIO_111	GRFC7_SEL
GPIO_30	CAM_I2C_SCL0	GPIO_71	CDC_PDM_TX	GPIO_112	GRFC8_SEL
GPIO_31	NC	GPIO_72	CDC_PDM_RX0	GPIO_113	GRFC9_SEL
GPIO_32	NC	GPIO_73	CDC_PDM_RX1	GPIO_114	GRFC10_SEL
GPIO_33	NC	GPIO_74	CDC_PDM_RX2	GPIO_115	GRFC11_SEL
GPIO_34	FLASH_STROBE_NOW	GPIO_75	BT_SSB1	GPIO_116	GRFC12_SEL
GPIO_35	MCAM1_PWDN	GPIO_76	WL_CMD_DATA_2	GPIO_117	NC
GPIO_36	NC	GPIO_77	WL_CMD_DATA_1	GPIO_118	EXT_GPS_LNA_EN
GPIO_37	FORCE_USB_BOOT	GPIO_78	WL_CMD_DATA_0	GPIO_119	CH0_GSM_TX_PHASE_D0
GPIO_38	NC	GPIO_79	WL_CMD_SET	GPIO_120	RFFE5_CLK
GPIO_39	SCAM_PWDN	GPIO_80	WL_CMD_CLK	GPIO_121	RFFE5_DATA
GPIO_40	SCAM_RST_N	GPIO_81	FM_SSB1	GPIO_122	NC

PMI8937 MPP Configuration	
MPP_1	NC
MPP_2	GREEN_LED
MPP_4	FLASH_STROBE_NOW

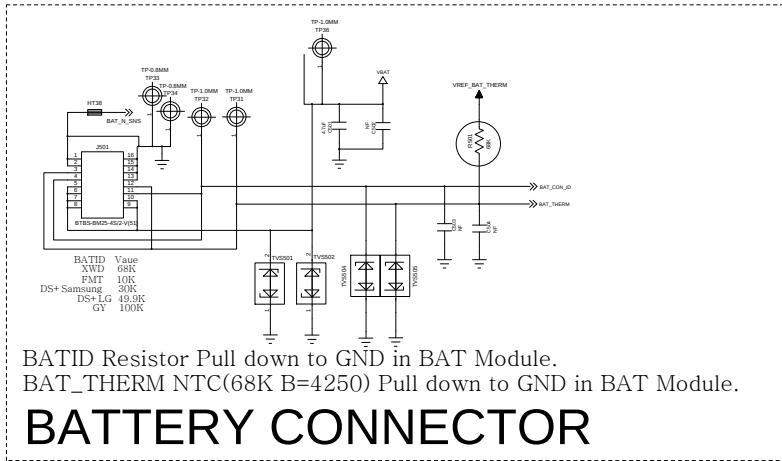
PM8937 GPIO/MPP Configuration			
GPIO_1	NC	MPP_1	VDD_PX_BIAS_MPP_1
GPIO_2	SDCARD_DET_N	MPP_2	NC
GPIO_3	UIM_BATT_ALM	MPP_3	VREF_DAC_MPP_3
GPIO_4	NC	MPP_4	QUIET_THERM
GPIO_5	NC		
GPIO_6	NC		
GPIO_7	NC		
GPIO_8	NC		

			Title	
			Street	
Size	Price			
Address			City	
State			Zip	

PMT_EN	connect to device without Wipower	connect to device with Wipower
CHG_OK	Pull up to VDD_CAP with 10k	Start's PMU CHG_OK
DVDD_EN	Pull down to GND with 10k	Start's PMU DVDD_EN
DPD2	No Connect	10k pull down to gnd
	For BATTERY CHARGING: Connect to GND in BAT	For BATTERY CHARGING: Connect to GND in pin. Do NOT pull up to VDD_CAP

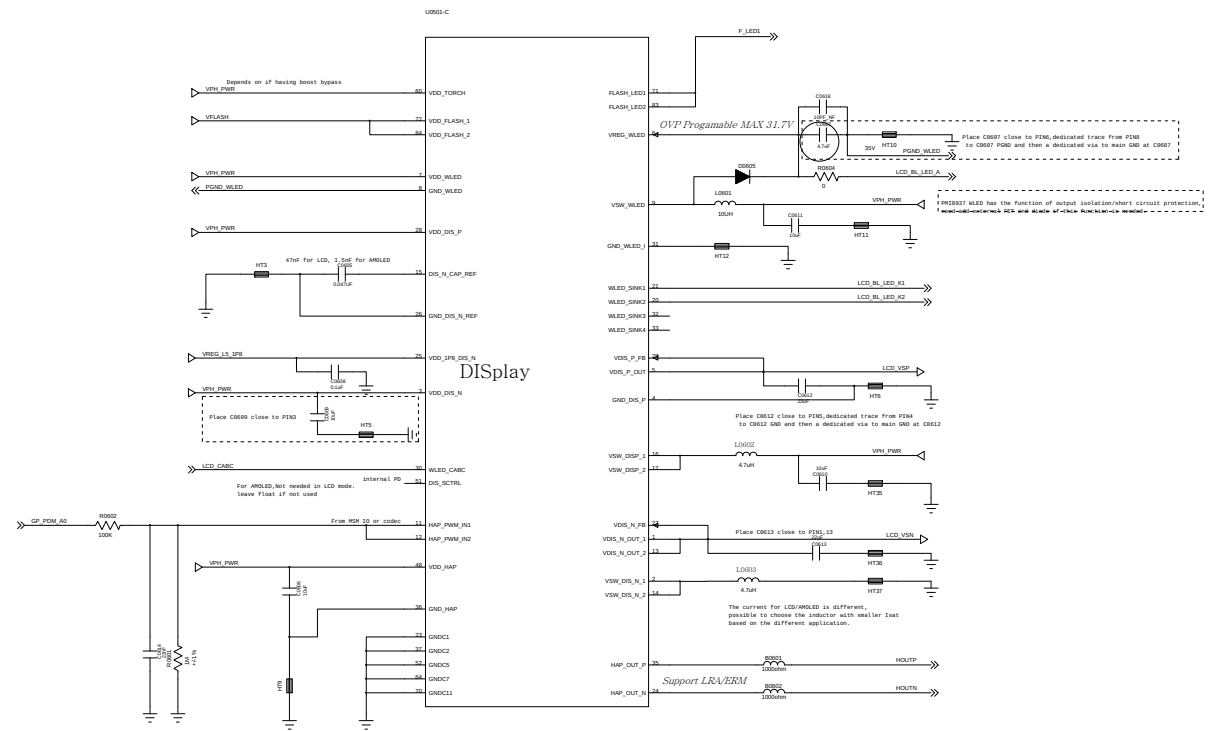


PMU VDD_CAP net is 5VDC. cap is 0603.
Do not use 0402 as may derate to ~80%.

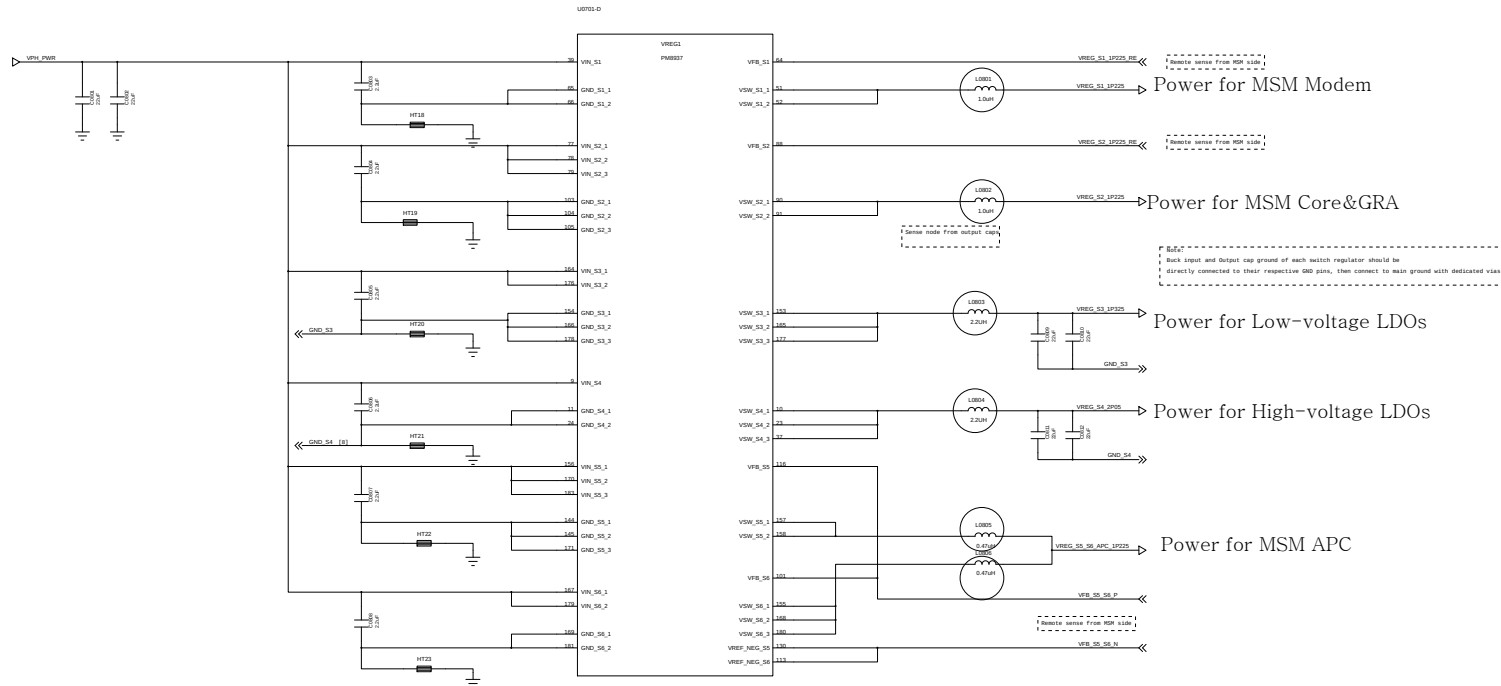


PMI8937 CHARGER

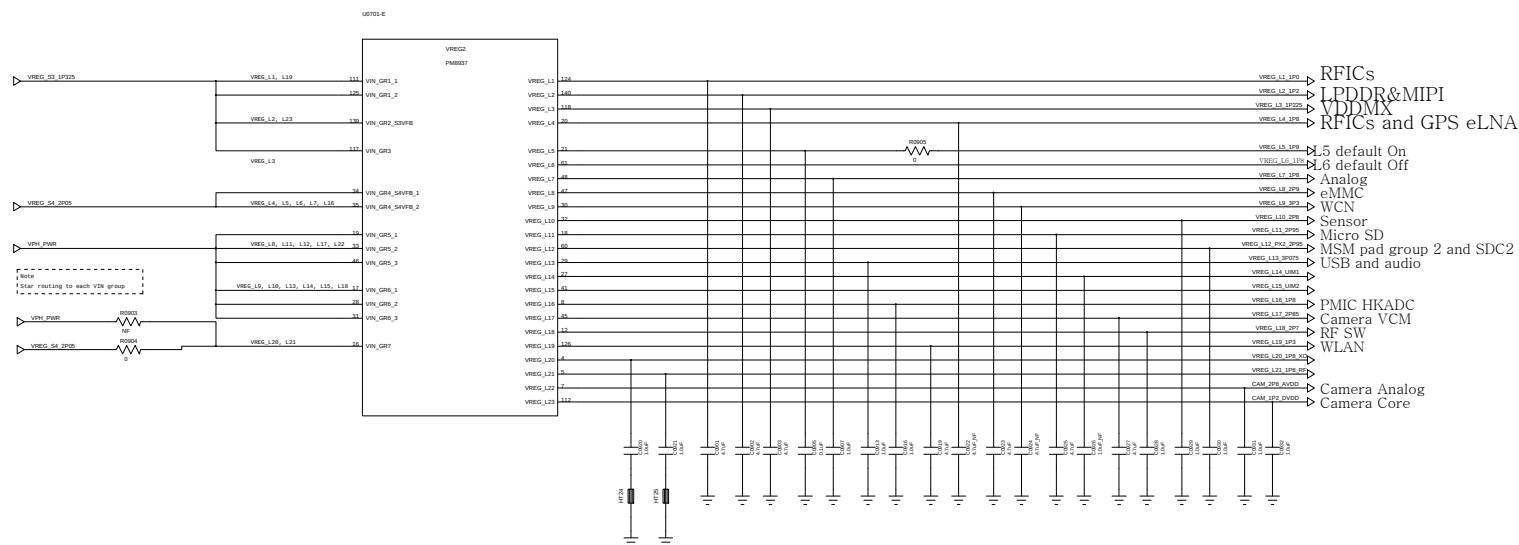
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Sheet	Rev	Rev
Size	Rev	Rev
Date	Rev	Rev



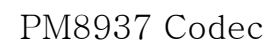
QUALCOMM Technologies, Inc. 3861 La Jolla Village Drive San Diego, CA 92121-1714 U.S.A.	
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Sheet Size Units	Rev
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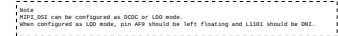
PM8937 Buck converter



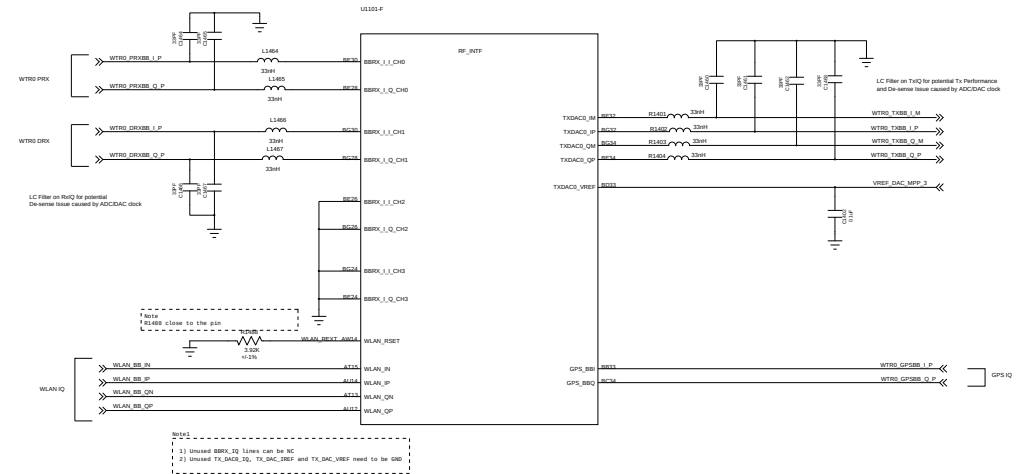
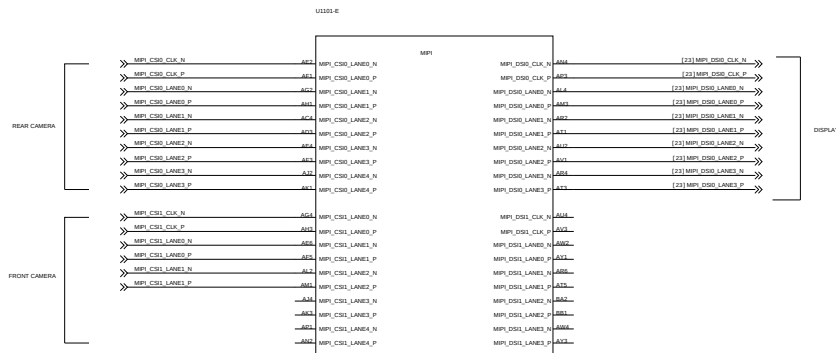
PM8937 LDO circuits



If eMMC 4.51 or earlier is used,
configure the unused MSM SDC1_RCLK pin to internal pull-down to prevent the bus from floating



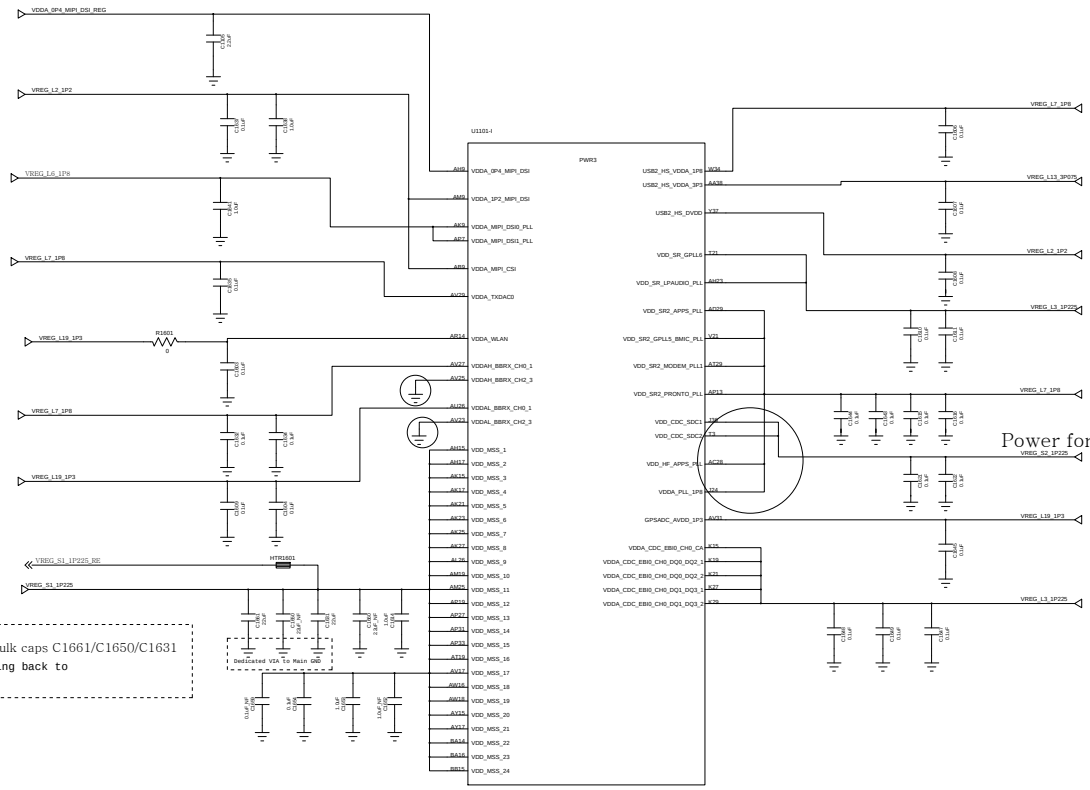
MSM8937 Control



Note: If best EMI practices are followed for MIPI CSI/DSI signals, there is no need for common mode choke filters. You may choose to have placeholders for common mode depending upon your design constraints. Extreme care must be taken that no stubs are created by doing so.

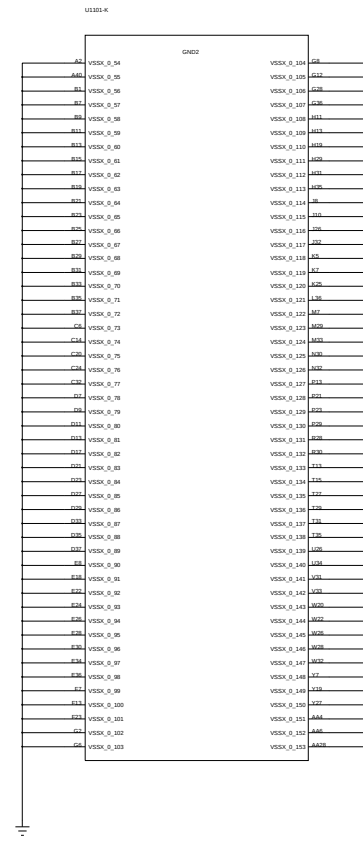
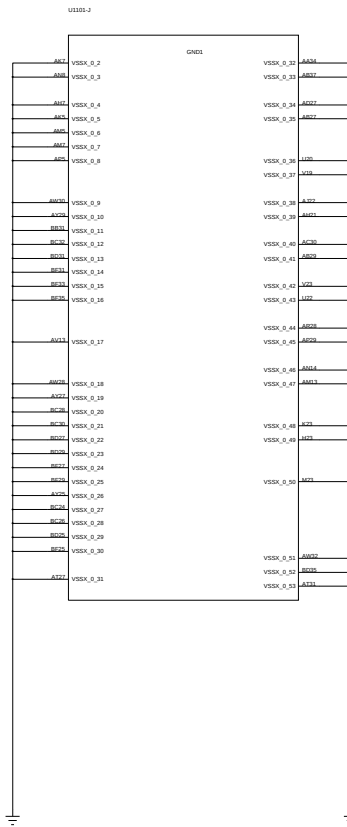
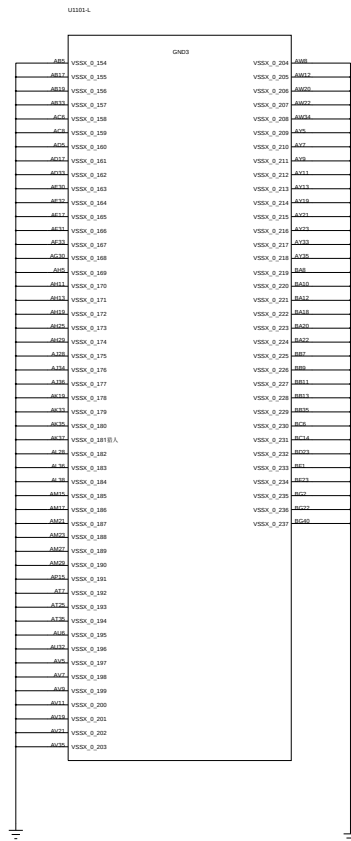
MSM8937 MIPI and RF Control

Note:
 HTR1601 are used to help show that the sense Bulk caps C1661/C1650/C1631
 are close to the MSM and the sense line feeding back to
 the PMIC should be placed after the caps.



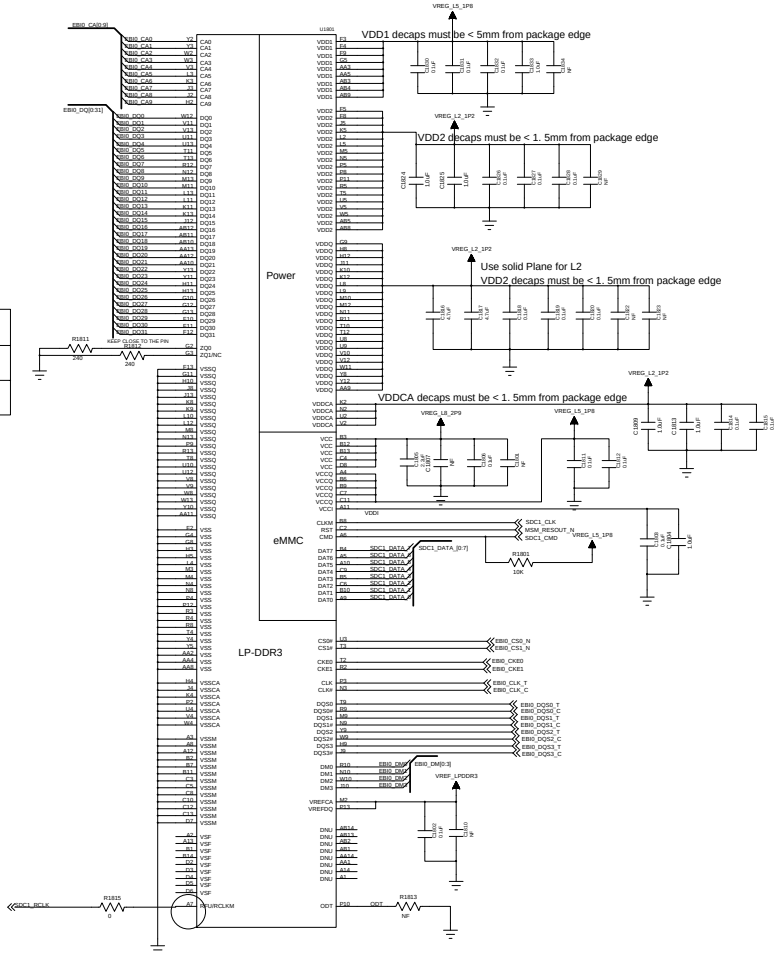
Power for secure digital calibration delay circuits

MSM8937 POWER2



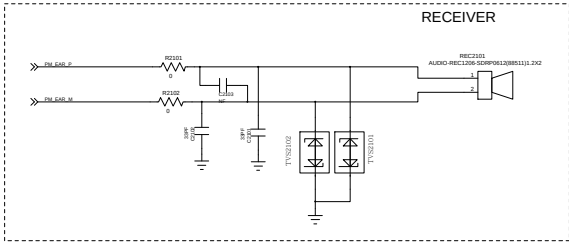
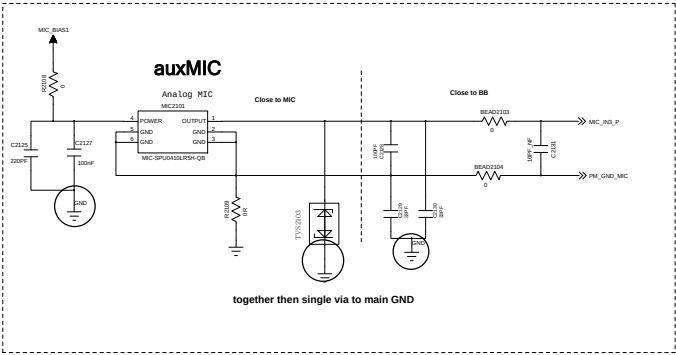
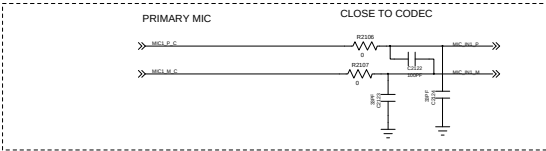
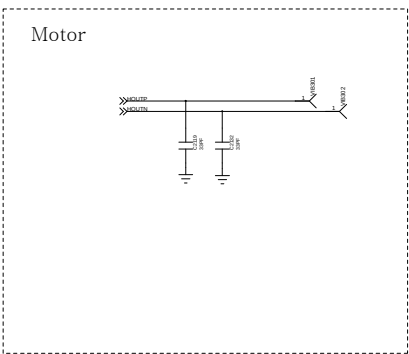
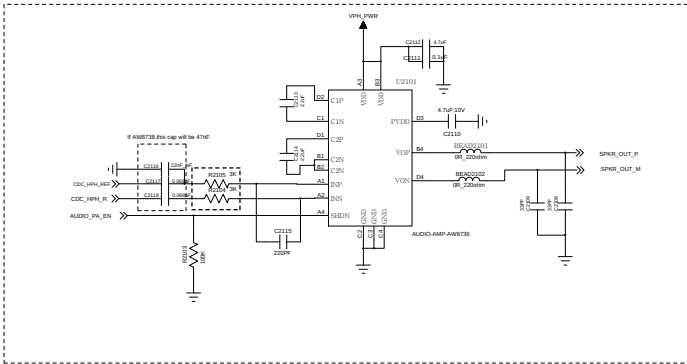
MSM8937 GND

	MICRO/HYNIX	SAMSUNG
U1501.G3	ZQ1	NC
U1501.A7	RFU	RCLK

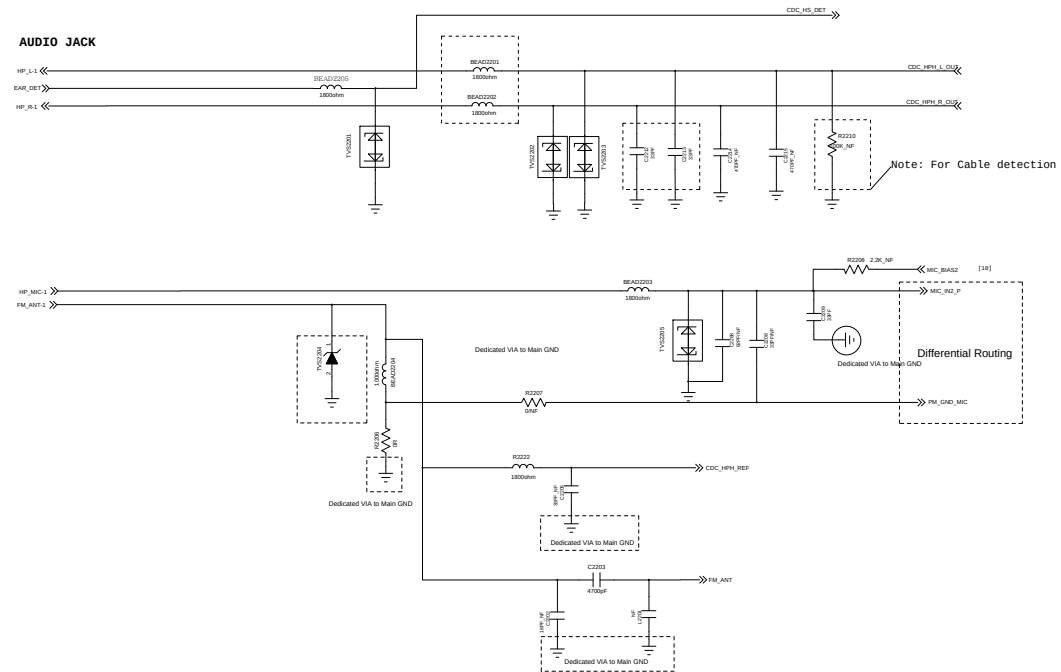
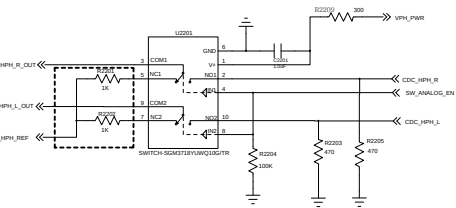


Note: Pull-up resistors on SDC1_DATA are PCB and eMMC vender dependent

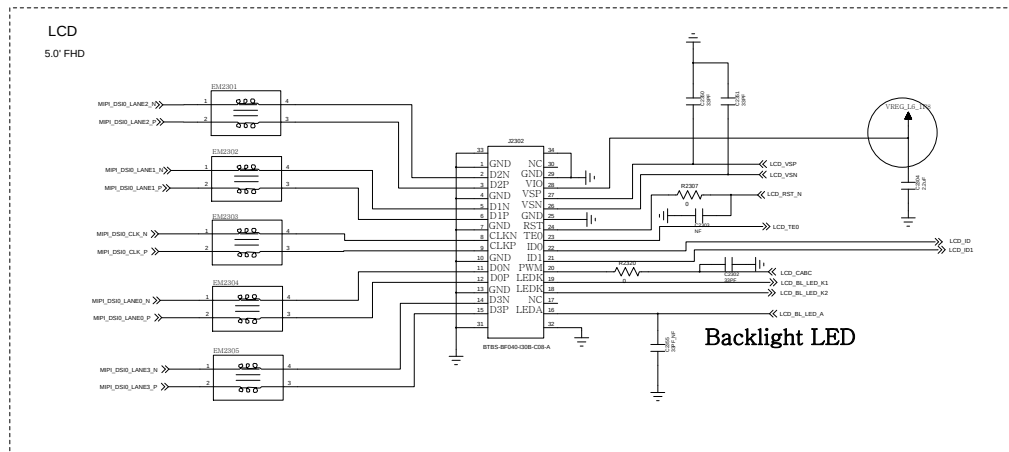
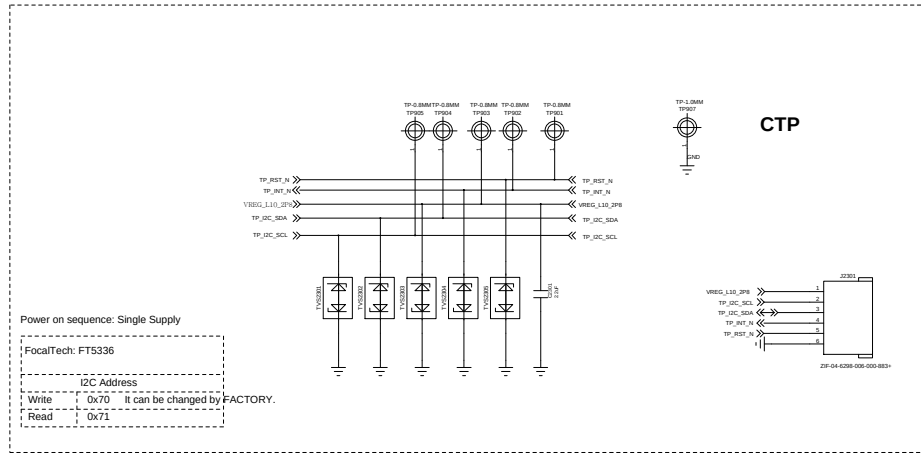
MEMORY:LPDDR3+ EMMC



SECONDARY MIC/RECEIVER



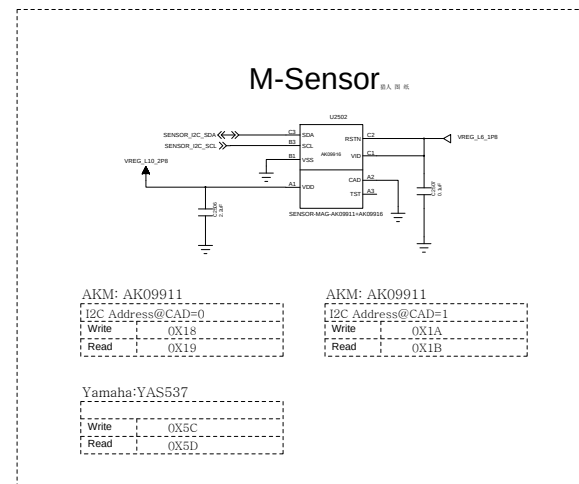
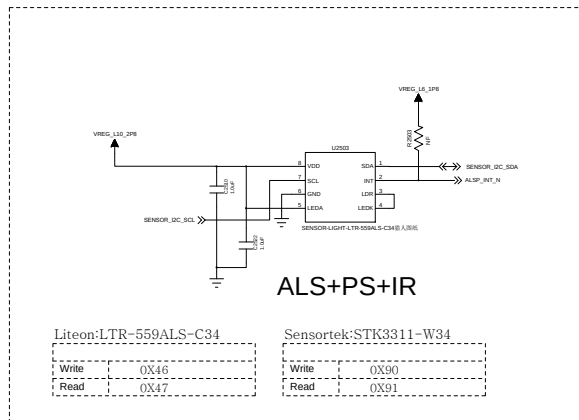
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Size	Name		Rev
Date		Sheet	of



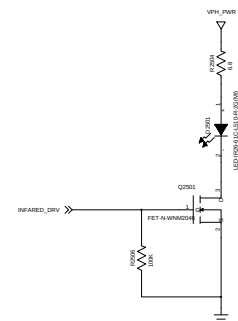
Note: If best EMI practices are followed for MIPI CSI/DSI signals, there is no need for common mode choke filters. You may choose to have placeholders for common mode depending upon your design constraints. Extreme care must be taken that no stubs are created by doing so.

LCD interface and backlight

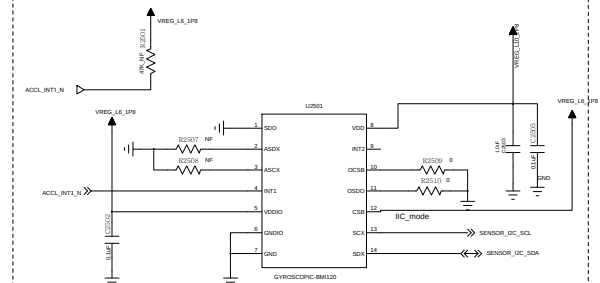
		QUALCOMM Technologies, Inc. 10000 Innovation Drive San Diego, CA 92121-1714 U.S.A.	
		Title Sheet	
		Size Date	
		Author Checked Date	



INFRARED LEDS



Gyroscope



Bosch:BMI120/BMI160

I2C Address@SDO=0

Write 0XD0

Read 0XD1

Bosch:BMI120/BMI160

I2C Address@SDO=1

Write 0XD2

Read 0XD3

ST:LSM6DS3TR

I2C Address@SDO=0

Write 0XD4

Read 0XD5

ST:LSM6DS3TR

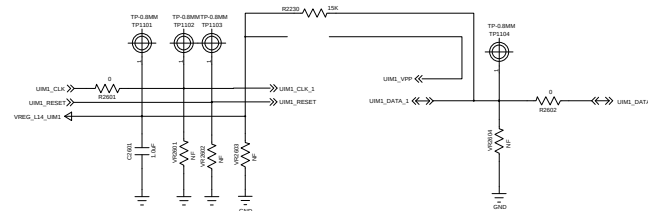
I2C Address@SDO=1

Write 0XD6

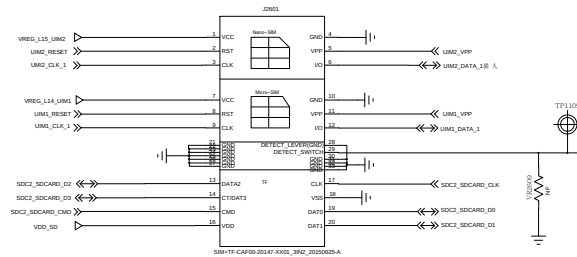
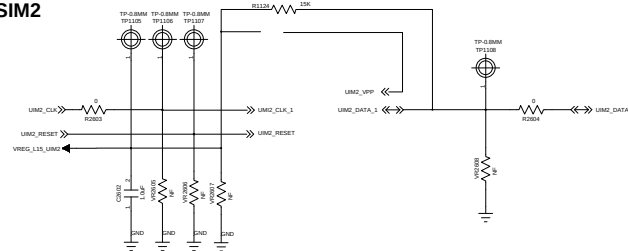
Read 0XD7

SENSORS

SIM1

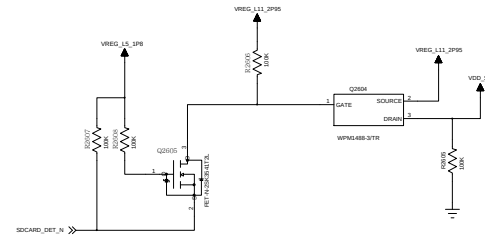


SIM2

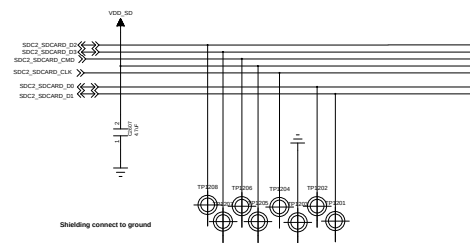


This int should give to GPIO54/GPIO58/GPIO67

Vout=3.0V,I-lim=1A

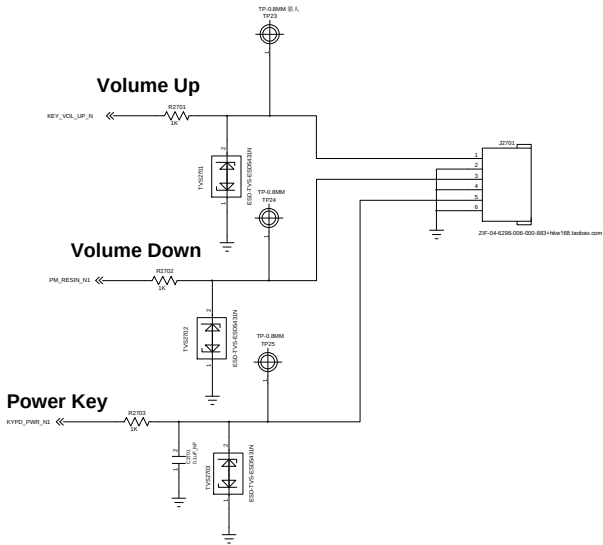


SD CARD

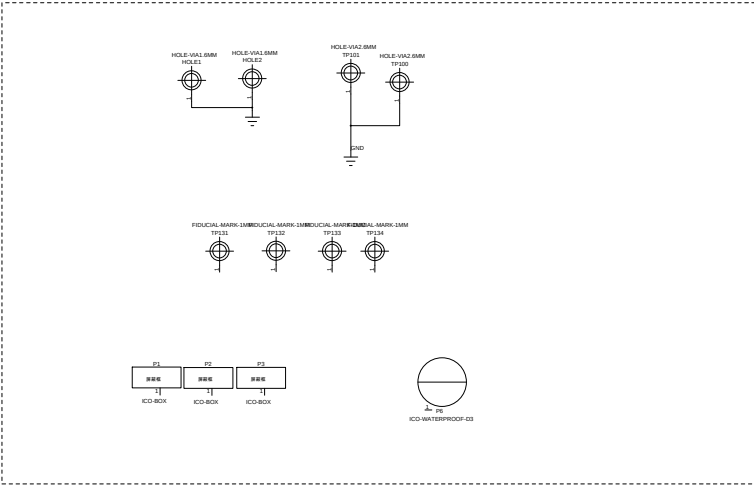


SIM/TF card

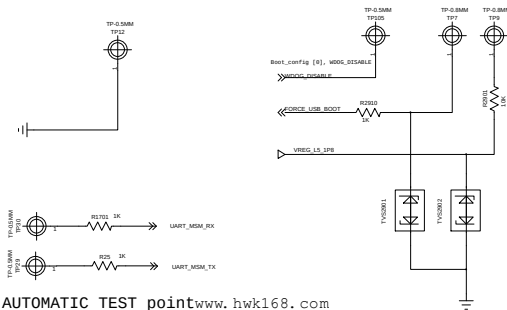
Signal	Description
KEY_VOL_UP_N	Volume Up
PM_RESIN_N1	Volume Down
KYPD_PWR_N1	Power ON
PM_RESIN_N1 + KYPD_PWR_N1	Hardware Reset



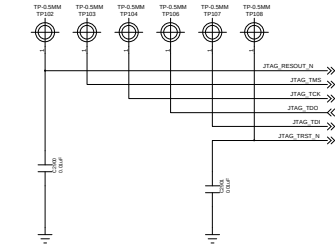
Keypad/LED/Status indicator



Test Point



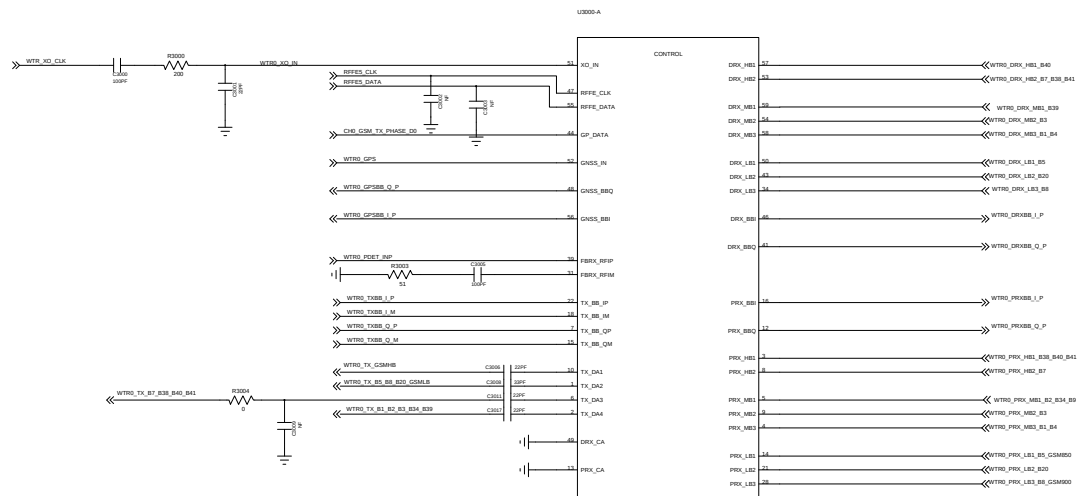
PCBA AUTOMATIC TEST pointwww.hw168.com



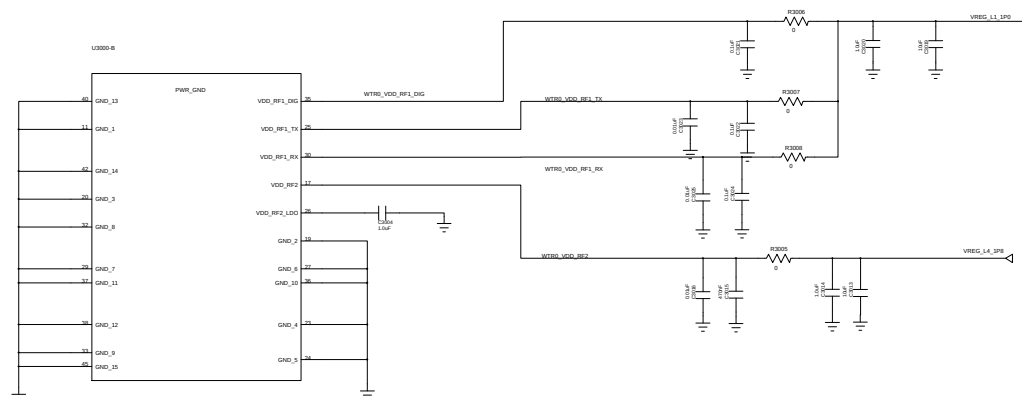
JTAG CONNECTOR

Test Point/Shields

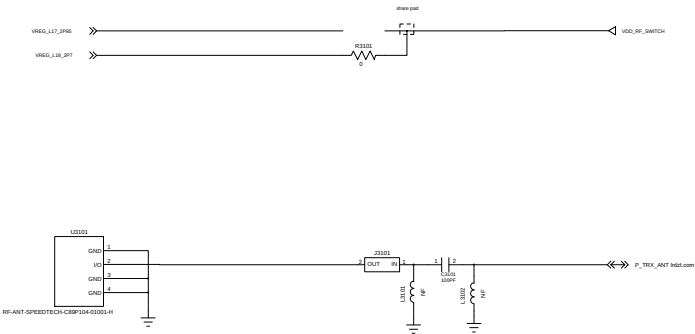
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Size		Date	
Rev		Rev	
Rev		Rev	



Note:RX ports have DC at the pin,
hwk168.com so it need DC
block,please make sure
there is no DC short to other voltages and GND



WTR 2955



Title					
Size		Sheet		Rev	
Date		Project		Of	

Band7

Band40

Band39

Band1

Band3

Band4

Band2

Band5

Band8

Band20

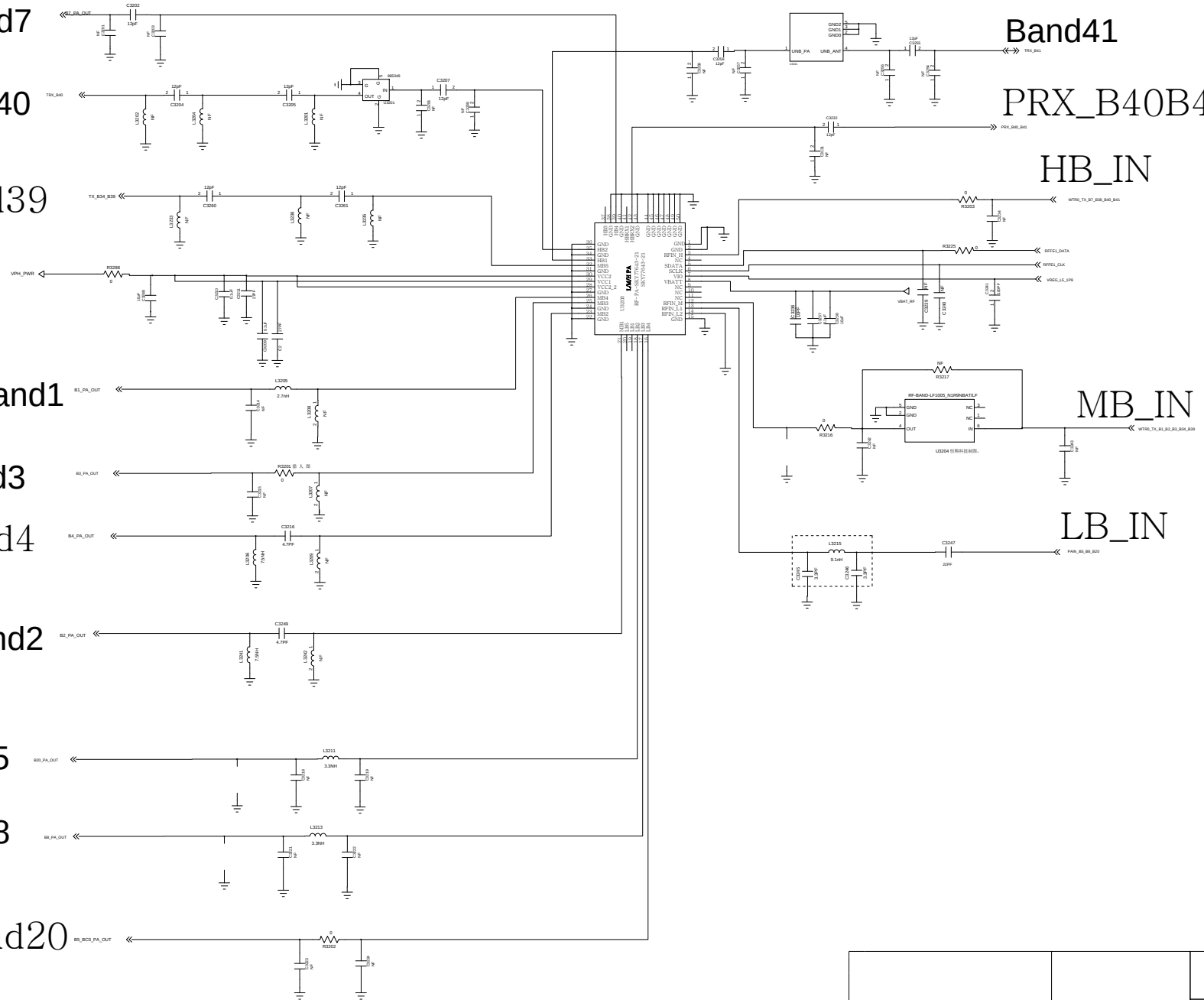
Band41

PRX_B40B41

HB_IN

MB_IN

LB_IN

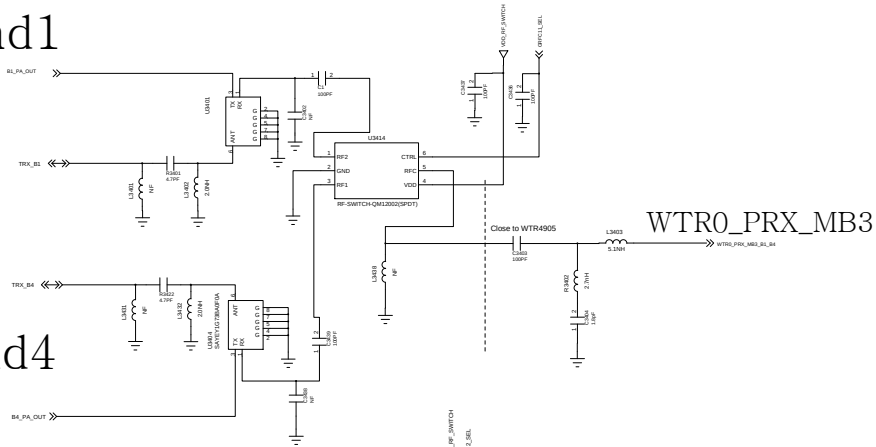




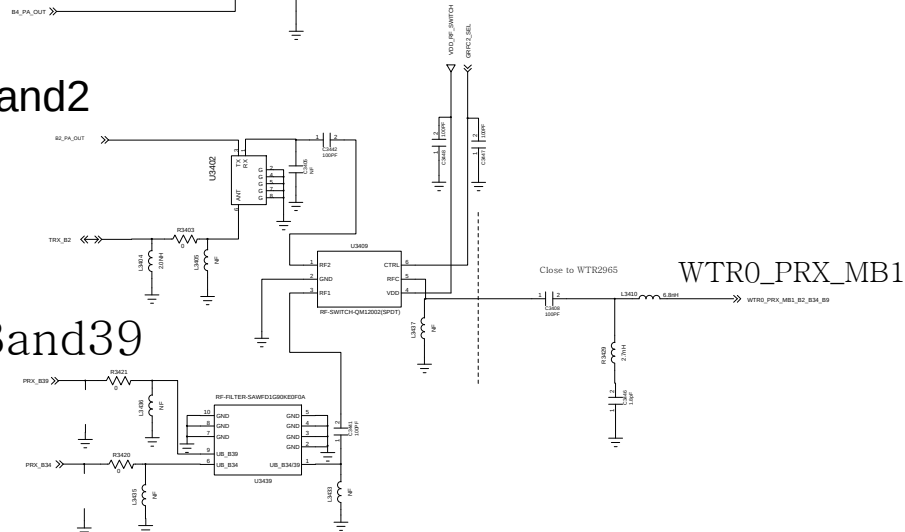
B5_B8_B20

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Sheet			
Size	Name		Rev
Date	Sheet		of

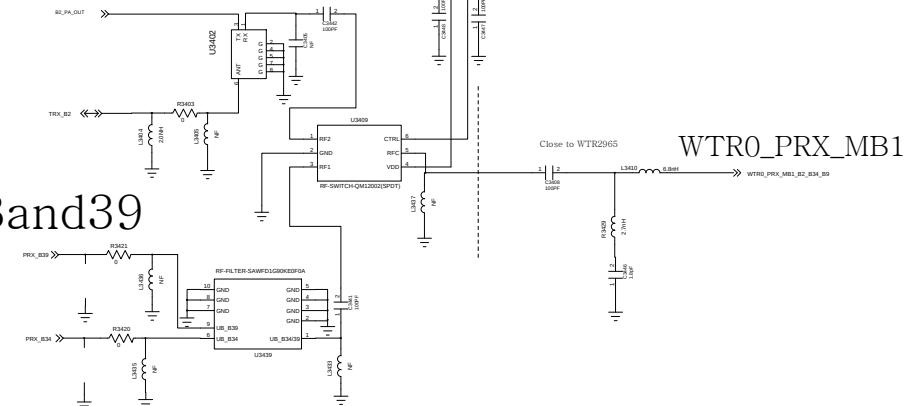
Band1



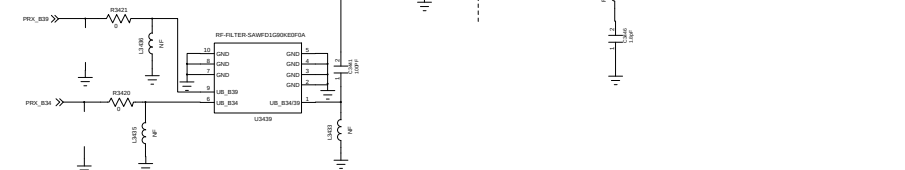
Band4



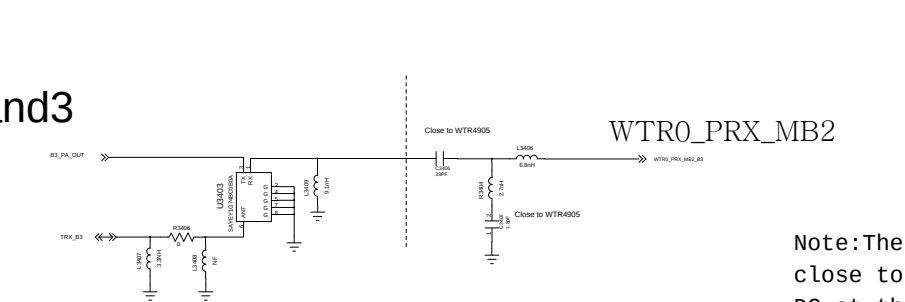
Band2



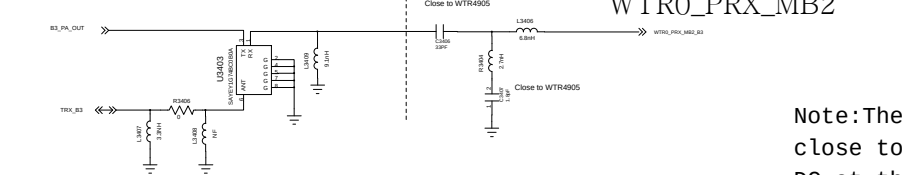
Band39



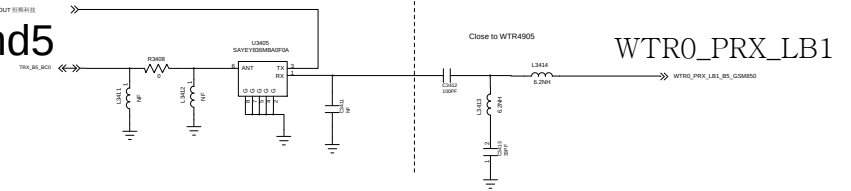
Band34



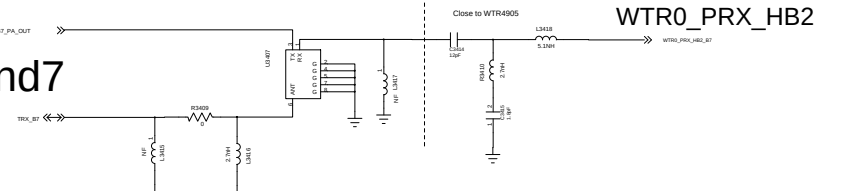
Band3



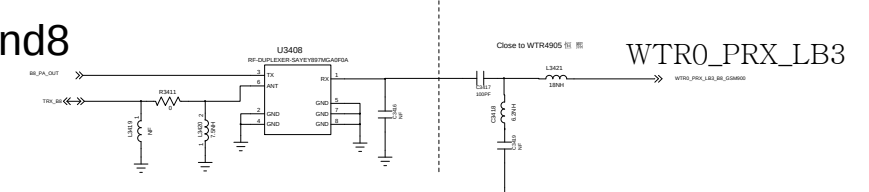
Band5



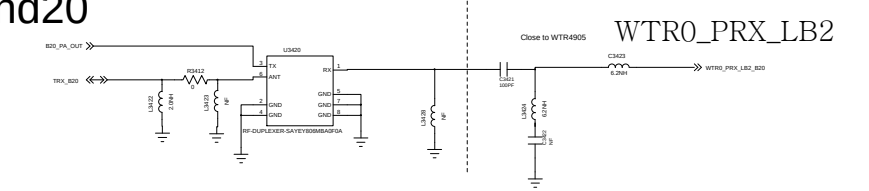
Band7



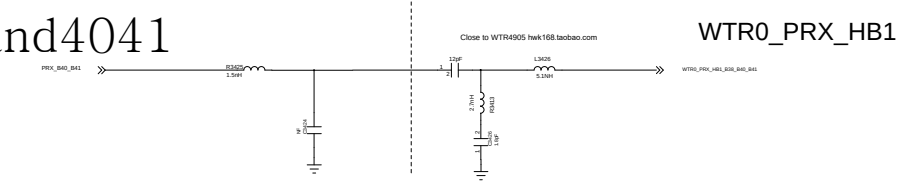
Band8



Band20

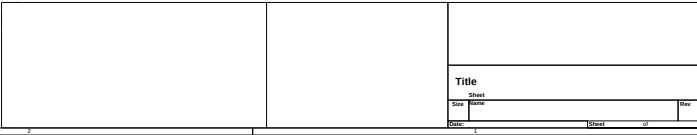


Band4041

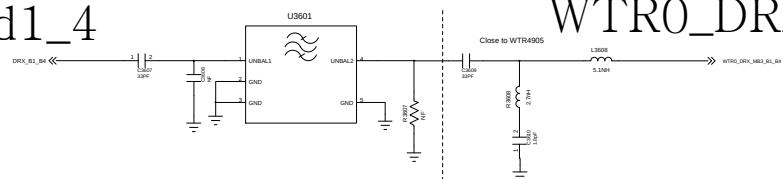


Note: The matching need close to WTR, RX ports have DC at the pin, so it need DC block,

Title	
Sheet	Rev
Date	Issue

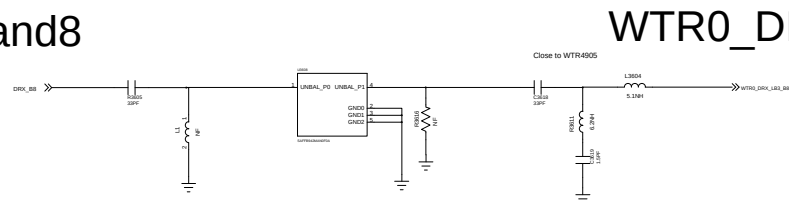


Band1_4



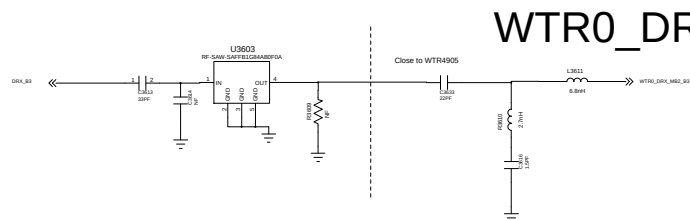
WTR0_DRX_MB3

Band8



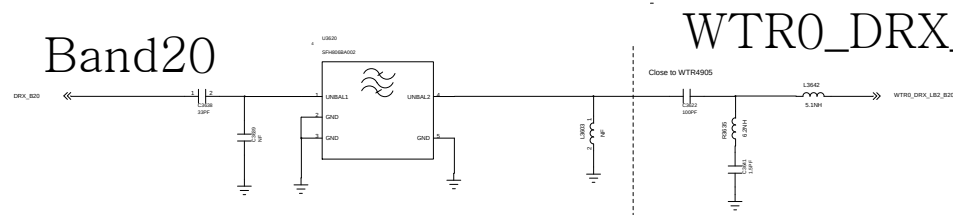
WTR0_DRX_LB3

Band3



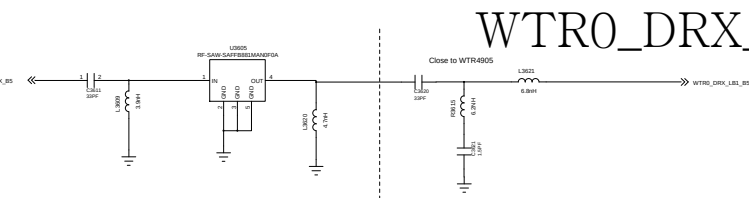
WTR0_DRX_MB2

Band20



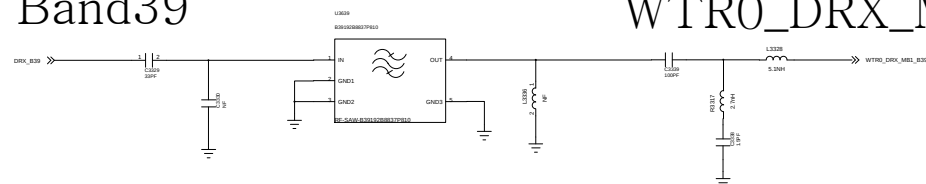
WTR0_DRX_LB2

Band5



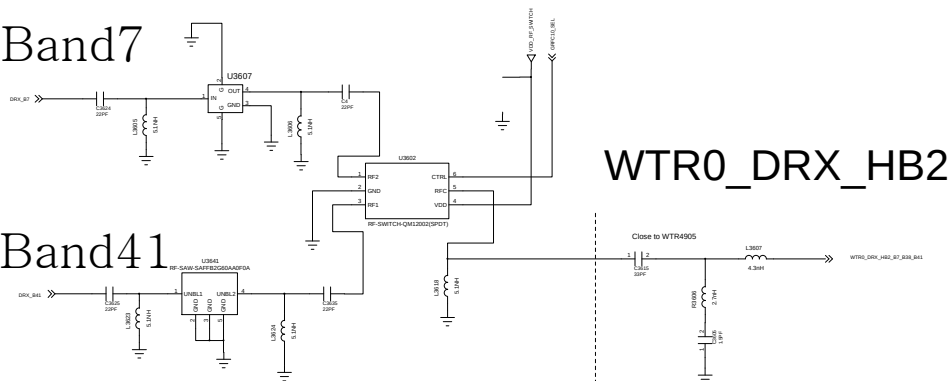
WTR0_DRX_LB1

Band39



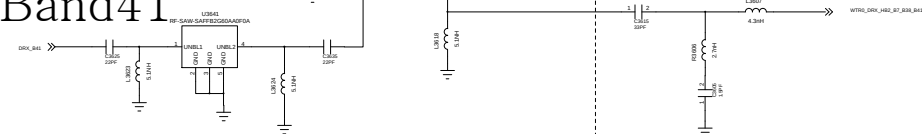
WTR0_DRX_MB1

Band7

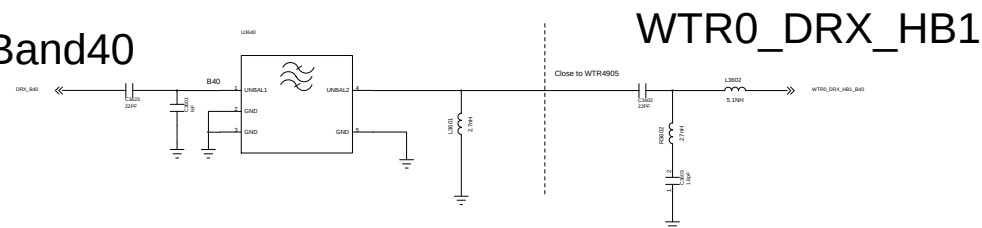


WTR0_DRX_HB2

Band41



Band40



WTR0_DRX_HB1

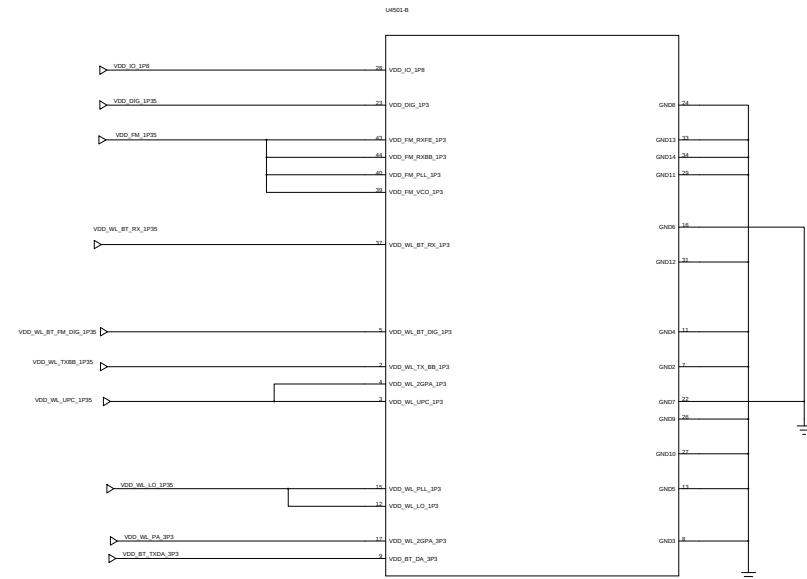
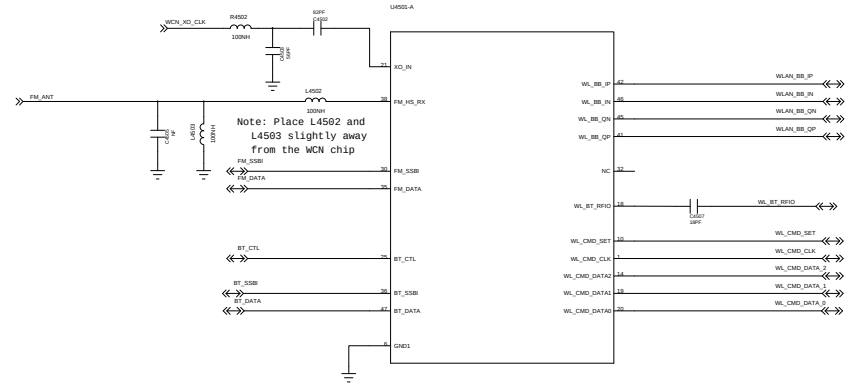
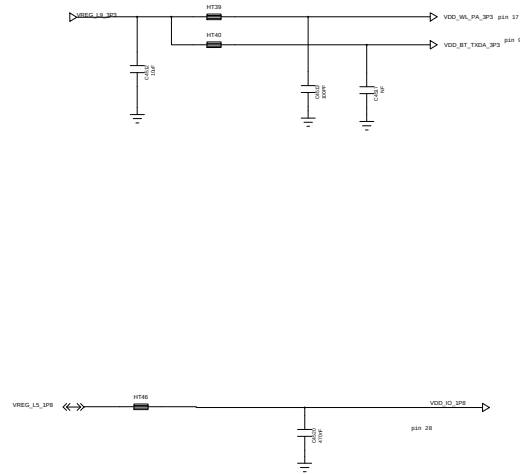
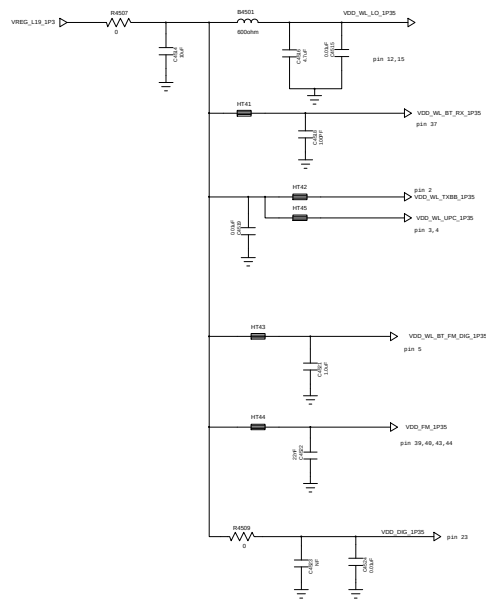
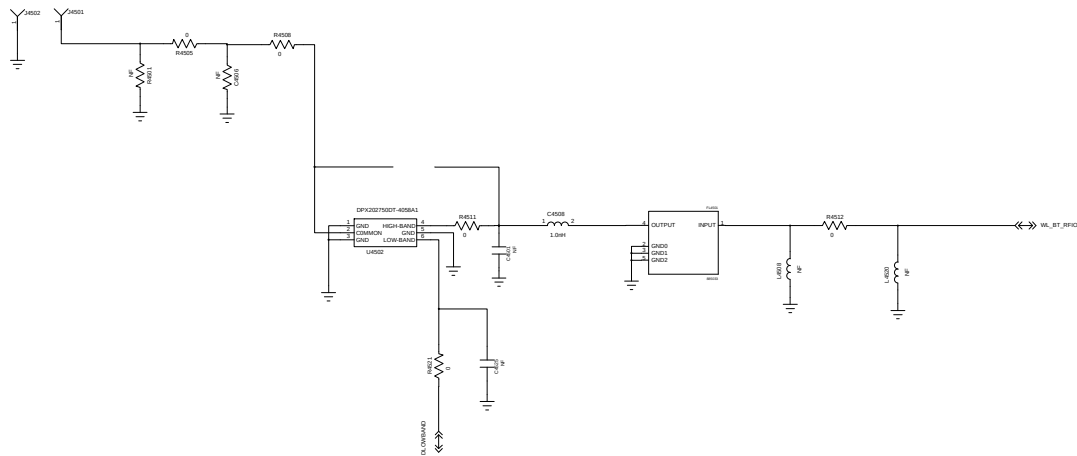
Note: The matching need close to WTR, RX ports have DC at the pin, so it need DC block,

Note: Use a single low impedance power plane/fill VPA for all ET PA VCC1/VCC2.

Refer to 80-NA681-91 rev.B or later revisions for more layout details.

				Title	
Size		Sheet		Rev	
Date		1 of 1			

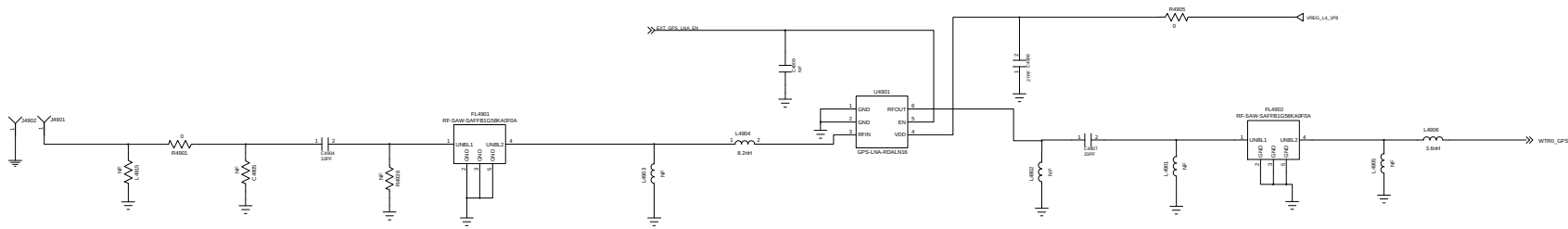
WCN



QUALCOMM Incorporated
Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

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Date
Rev

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